Using this information, estimate the following quantities (a) total amount of charge transferred between cloud and ground (b) the current in the lightning bolt (c) the power delivered in 0.2 s.

Ans: charge = 20 C, I = 100 A, P = 5 GW

3. A copper wire of 10^{-6} m^2 area of cross section, carries a current of 2 A. If the number of free electrons per cubic meter in the wire is 8×10^{28}, calculate the current density and average drift velocity of electrons.

Ans: J = 2×10^{6} Am^{-2}

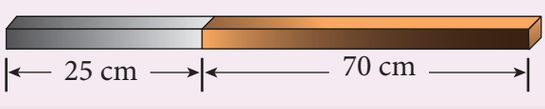
$v_d = 15.6 \times 10^{-5}$ ms^{-1}

4. The resistance of a nichrome wire at 20^{0}C is 10 Ω. If its temperature coefficient of resistivity of nichrom is 0.004/^{0}C, find the resistance of the wire at boiling point of water. Comment on the result.

Ans: R_T = 13.2 Ω.

As the temperature increases the resistance of the wire also increases.

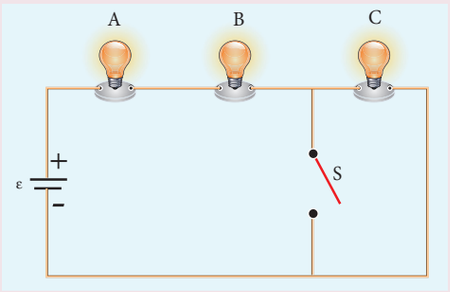
5. The rod given in the figure is made up of two different materials.



Both have square cross sections of 3 mm side. The resistivity of the first material is 4×10^{-3} Ωm and that of second material has resistivity of 5×10^{-3} Ωm. What is the resistance of rod between its ends?

Ans: 500 Ω

6. Three identical lamps each having a resistance R are connected to the battery of emf ε as shown in the figure.



Suddenly the switch S is closed. (a) Calculate the current in the circuit when S is open and closed (b) What happens to the intensities of the bulbs A,B and C. (c) Calculate the voltage across the three bulbs when S is open and closed (d) Calculate the power delivered to the circuit when S is opened and closed (e) Does the power delivered to the circuit decrease, increase or remain same?

Ans:

Electrical quantities	Switch S is open	Switch S is closed
a) Current	$\frac{\varepsilon}{3R}$	$\frac{\varepsilon}{2R}$
b) Voltage	$V_A = \frac{\varepsilon}{3}$ $V_B = \frac{\varepsilon}{3}$ $V_C = \frac{\varepsilon}{3}$	$V_A = \frac{\varepsilon}{2}$ $V_B = \frac{\varepsilon}{2}$ $V_C = 0$
c) Power	$P_A = \frac{\varepsilon^2}{9R}$ $P_B = \frac{\varepsilon^2}{9R}$ $P_C = \frac{\varepsilon^2}{9R}$	$P_A = \frac{\varepsilon^2}{4R}$ $P_B = \frac{\varepsilon^2}{4R}$ $P_C = 0$ Total power increases
d) Intensity	All the bulbs glow with equal intensity	The intensities of the bulbs A and B equally increase. Bulb C will not glow since no current pass through it.
e) Increases		

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